

First-order Intuitionistic Multiplicative Exponential Linear Logic with Hilbert's Epsilon

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Introduction

- The Epsilon Calculus: introduced by Hilbert to study proofs and consistency of logical theories.
- More recently, a renewed interest for its interesting properties in classical first-order proof theory, such as compression of cut-free proofs.
- This (ongoing) work aims to study in what capacity the ε -calculus can be integrated into Linear Logic, as a novel approach to LL quantification.

1 Hilbert's ε -calculus

- The ε -operator
- The ε -theorems
- Proof compression

2 IMELL ε -calculus

3 Categorical semantics

- Hyperdoctrines
- ε -hyperdoctrines
- IMELL ε -hyperdoctrines

4 Further research

- Semantics for classical MELL
- Proof nets ?

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$$A(t) \rightarrow A(\varepsilon_x A)$$

or in sequent-style:

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 $\leadsto$ (barely) disguised injection of the Axiom of Choice in the syntax.

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First ε -theorem

If E is a formula without ε -terms, and if there is an ε -proof of E , then there is a proof of E containing no critical ε -rule.

$\leadsto$ Conservativity over the elementary calculus.

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Extended First ε -theorem

If $E(e_1, \dots, e_m)$ is a (quantifier-free) formula containing only the ε -terms $e_1, \dots, e_m$, and if there is a proof of $E(e_1, \dots, e_m)$, then there are ε -free terms t_j^i , $1 \leq i \leq n, 1 \leq j \leq m$ and a proof of $\bigvee_{i=1}^n E(t_1^i, \dots, t_m^i)$ containing no critical ε -rule.

$\leadsto$ Analogue to Herbrand's Theorem for ε -calculi.

Non-elementary proof compression

Proof compression

- The ε -calculus admits shorter cut-free proofs than traditional sequent calculus.

Baaz, Lolić, 2024

There are sequents for which the size of their smallest cut-free LK proof is not bounded by any elementary function of the size of their smallest cut-free ε -proof.

- Allett, 2024: similar result on non-elementary compression of deep inference proofs.

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- Example: shortest cut-free proof of the Drinkers' Formula

$$\begin{array}{c}
 \text{ax} \frac{}{\vdash \neg P(t), P(z), \neg P(z), \forall y P(y)} \\
 \vee \frac{}{\vdash \neg P(t), P(z), \neg P(z) \vee \forall y P(y)} \\
 \exists \frac{}{\vdash \neg P(t), P(z), \exists x (\neg P(x) \vee \forall y P(y))} \\
 \vee \frac{}{\vdash \neg P(t), \forall y P(y), \exists x (\neg P(x) \vee \forall y P(y))} \\
 \vee \frac{}{\vdash \neg P(t) \vee \forall y P(y), \exists x (\neg P(x) \vee \forall y P(y))} \\
 \exists \frac{}{\vdash \exists x (\neg P(x) \vee \forall y P(y)), \exists x (\neg P(x) \vee \forall y P(y))} \\
 \text{cont} \frac{}{\vdash \exists x (\neg P(x) \vee \forall y P(y))}
 \end{array}$$

$$\begin{array}{c}
 \text{ax} \frac{}{\vdash \neg P(\varepsilon_y \neg P) \vee P(\varepsilon_y \neg P)} \\
 \varepsilon R \frac{}{\vdash \neg P(\varepsilon_x (\neg P(x) \vee P(\varepsilon_y \neg P)) \vee P(\varepsilon_y \neg P))}
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Linear Logic with ε

- Several reasons to consider adding ε to Linear Logic:
 - Leverage the shorter proofs provided by the ε -calculus.
 - Treat quantification in multiplicative LL without the inherently additive $\forall, \exists$.
 - Also a way to get rid of quantifier boxes in proof nets (more on that later).

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 - Leverage the shorter proofs provided by the ε -calculus.
 - Treat quantification in multiplicative LL without the inherently additive $\forall, \exists$.
 - Also a way to get rid of quantifier boxes in proof nets (more on that later).
- For now, we restrict ourselves to Intuitionistic Multiplicative (Exponential) Linear Logic (IM(E)LL).

- IMELL ε terms t and formulae $\mathcal{F}$ are defined conjointly:

$$t := x \mid c \mid f(t, \dots, t) \mid \varepsilon_x \mathcal{F}$$

$$\mathcal{F} := \mathbf{1} \mid \perp \mid P(t, \dots, t) \mid \mathcal{F} \otimes \mathcal{F} \mid \mathcal{F} \multimap \mathcal{F} \mid !\mathcal{F}$$

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- Sequent calculus: same rules as traditional IMELL, but quantifier rules are replaced by ε -rules.

$$\varepsilon_L \frac{\Gamma, A(x) \vdash C}{\Gamma, A(\varepsilon_x A) \vdash C} \quad x \notin \text{free}(\Gamma) \qquad \varepsilon_R \frac{\Gamma \vdash A(t)}{\Gamma \vdash A(\varepsilon_x A)}$$

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- Usual semantics of the classical ε -calculus: classical models with choice functions.

$$\mathcal{M}, \varphi \models A(\varepsilon_x B) \quad \text{iff} \quad \varphi(\llbracket B \rrbracket^{\mathcal{M}}) \in \llbracket A \rrbracket^{\mathcal{M}}$$

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- Instead, we use categorical semantics in the form of hyperdoctrines.

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First-order Hyperdoctrine

A **first-order hyperdoctrine** is a couple (C, P) where:

- C is a category with finite products.
- $P : C^{op} \rightarrow \mathbf{Pos}$ is a contravariant functor to the category of posets.

Given a first-order language $\mathcal{L}$, we can define the category $Ctx_{\mathcal{L}}$ of $\mathcal{L}$ -contexts, in which:

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- The identity of $\vec{x}$ is the identity substitution $[x_1/x_1, \dots, x_n/x_n]$.

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If $P : \text{Ctx}_{\mathcal{L}}^{\text{op}} \rightarrow \mathbf{Pos}$ is a contravariant functor, then $(\text{Ctx}_{\mathcal{L}}, P)$ is a hyperdoctrine:

- Products and projections in $\text{Ctx}_{\mathcal{L}}$ are context products/projections.
- Each $P(\vec{x})$ can be seen as the collection of $\mathcal{L}$ -formulae over the context $\vec{x}$, ordered by entailment: $\phi(\vec{x}) \leq \psi(\vec{x})$ iff $\phi(\vec{x}) \vdash \psi(\vec{x})$ is provable.

$$\begin{array}{ccc} \vec{x} \times y & \xrightarrow{\pi} & \vec{x} \\ \downarrow P & & \downarrow P \\ P(\vec{x} \times y) & \xleftarrow{P(\pi)} & P(\vec{x}) \end{array}$$

Hyperdoctrines

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$(\text{Ctx}_{\mathcal{L}}, P)$ is said to be **existential** (resp. **universal**) if for every projection π , $P(\pi)$ has a left adjoint $\exists\pi$ (resp. right adjoint $\forall\pi$):

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 \downarrow P & & \downarrow P \\
 P(\vec{x} \times y) & \begin{array}{c} \xleftarrow{\exists\pi} \\ \xrightarrow{P(\pi)} \end{array} & P(\vec{x}) \\
 & \text{\scriptsize $\forall\pi$} &
 \end{array}$$

such that:

- Quantification commutes with substitution (**Beck-Chevalley condition**)
- For every $\phi \in P(\vec{x} \times y)$ and $\psi \in P(\vec{x})$, we have: $\exists\pi(P(\pi)(\psi) \wedge \phi) = \exists\pi(\phi) \wedge \psi$ (**Frobenius reciprocity**).

Additional structure of the posets provide categorical semantics for various first-order logics, depending on the internal logic of each $P(\vec{x})$ (when seen as categories):

- Heyting algebras: Intuitionistic first-order logic.
- Boolean algebras: Classical first-order logic.
- Star-autonomous categories: First-order Multiplicative Linear Logic.
- etc.

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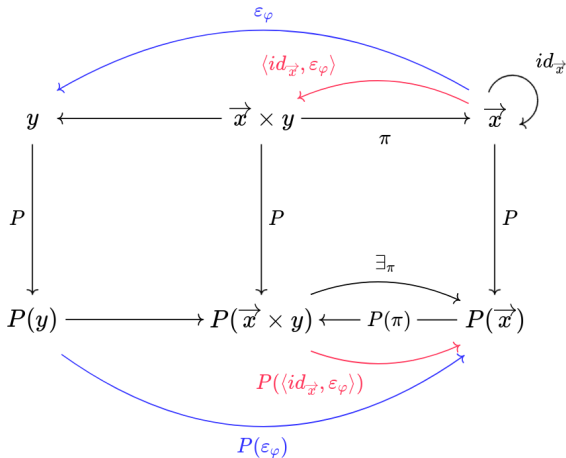
ε -hyperdoctrine

($\text{Ctx}_{\mathcal{L}}, P$) is said to have the ε -operator if for every $\vec{x}, y$ and every $\varphi \in P(\vec{x} \times y)$, there is a morphism $\varepsilon_{\varphi} : \vec{x} \rightarrow y$ such that:

$$\exists_{\pi}(\varphi) = P(\langle id_{\vec{x}}, \varepsilon_{\varphi} \rangle)(\varphi)$$

ε -hyperdoctrines

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$$\exists_\pi(\varphi) = P(\langle id_{\vec{x}}, \varepsilon_\varphi \rangle)(\varphi) \quad \text{means} \quad \exists x \varphi(x) \equiv \varphi(\varepsilon_x \varphi(x))$$

IMELL ε -hyperdoctrines

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- An interpretation of the exponential " $!$ ", i.e. a comonad on $P(\vec{x})$.

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- An interpretation of the exponential "!", i.e. a comonad on $P(\vec{x})$.
- And an interpretation of ε -terms.

So we just have to find the right structure for the truth values/interpretations of formulae in $P(\vec{x})$.

Truth value ordered monoid

IMELL ε -hyperdoctrines

- Define $M = (\{0\} \cup \{\frac{1}{2^n} \mid n \in \mathbb{N}\}, \cdot, \leq)$, with $\cdot, \leq$ the standard multiplication and order on $\mathbb{Q}$.

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 - $\llbracket A \multimap B \rrbracket = \begin{cases} 1 & \text{if } \llbracket A \rrbracket = 0 \\ \min(1, \llbracket B \rrbracket / \llbracket A \rrbracket) & \text{otherwise} \end{cases}$

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- " $!$ " is a comonad, so we need:
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- But then the interpretation of " $!$ " is too trivial.

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- This leaves one possibility: $\llbracket !A \rrbracket = \begin{cases} 1 & \text{if } \llbracket A \rrbracket = 1 \\ 0 & \text{otherwise} \end{cases}$

- But then the interpretation of " $!$ " is too trivial.

- We simply take M^ω , ω copies of M for truth values, with $\cdot, \leq, !$ defined pointwise on M^ω .

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$$\llbracket \exists_{\pi}(A) \rrbracket \leq \llbracket P(\langle id_{\overline{X}}, \varepsilon_A \rangle)(A) \rrbracket$$

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- Every non-empty subset of M^{ω} has a maximal element, so we simply take:

$$\llbracket P(\langle id_{\vec{X}}, \varepsilon_A \rangle)(A) \rrbracket = \max_{\pi}(\llbracket \exists_{\pi}(A) \rrbracket)$$

IMELL ε -hyperdoctrines

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In summary:

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- Every ε -hyperdoctrine in which for each poset $P(\vec{x})$, the formulae over the context $\vec{x}$ are assigned truth values in the ordered monoid M^ω , provides an interpretation for IMELL $+ \varepsilon$.
- The semantics proposed are semantics of *formulae*, not semantics of *proofs*.
- We work with intuitionistic linear logic to avoid issues related to negation, double-negation closure, etc.

1 Hilbert's ε -calculus

- The ε -operator
- The ε -theorems
- Proof compression

2 IMELL ε -calculus

3 Categorical semantics

- Hyperdoctrines
- ε -hyperdoctrines
- IMELL ε -hyperdoctrines

4 Further research

- Semantics for classical MELL
- Proof nets ?

Classical MELL + ε

Semantics for classical MELL

- As defined here, IMELL ε -hyperdoctrines do not accomodate for linear negation $(.)^\perp$, and thus for universal quantification in the form $\forall x A(x) \equiv A(\varepsilon_x A^\perp)$.

Classical MELL + ε

Semantics for classical MELL

- As defined here, IMELL ε -hyperdoctrines do not accommodate for linear negation $(.)^\perp$, and thus for universal quantification in the form $\forall x A(x) \equiv A(\varepsilon_x A^\perp)$.
- One could define $A^\perp$ as $A \multimap \perp$, but in this case this is not fine-grained enough, since $A^\perp$ always collapses to $\perp$ unless $\llbracket A \rrbracket = 0$.
- Future investigations: provide similar categorical semantics for classical MELL + ε .

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- It also eliminates the need for quantifier rules with eigenvariables, leaving only the ε -rule as sole quantification rule:

$$\varepsilon \frac{\Gamma \vdash \Delta, A(t)}{\Gamma \vdash \Delta, A(\varepsilon_x A)}$$

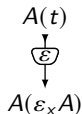
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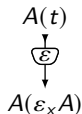
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- No eigenvariable conditions $\leadsto$ no quantifier boxes.

Thank you for your attention !